Three Approaches to AI

Standard AI vs Simulated Neuromorphic AI vs Neuromorphic AI on a Neuromorphic Chip

THINK OF THESE AS THREE LAYERS OF THE STACK

1. **Standard AI** = conventional software + conventional hardware.
2. **Simulated neuromorphic AI** = brain-inspired software running on conventional hardware.
3. **Neuromorphic AI on neuromorphic chips** = brain-inspired software running on brain-inspired hardware.

CONCEPTUAL BRAIN-LIKENESS — ILLUSTRATIVE, NOT A SCIENTIFIC MEASUREMENT

Standard AI
2 / 10

Simulated Neuromorphic
6 / 10

Neuromorphic on chip
10 / 10

Conceptual comparison, not a benchmark.

ASPECT BY ASPECT

Aspect	1 · Standard AI	2 · Simulated Neuromorphic	3 · Neuromorphic on Chip
HARDWARE	CPUs, GPUs, TPUs	CPUs, GPUs, TPUs	Specialized neuromorphic processors (e.g., Loihi, TrueNorth, Akida)
COMPUTATION	Matrix multiplication	Brain-inspired neural dynamics simulated in software	Native spike-based computation
NEURONS	Artificial neurons	Simulated spiking neurons	Hardware spiking neurons
TIMING	Synchronous (clock-driven)	Often event-driven	Event-driven (asynchronous)
ENERGY USE	High	Moderate to high	Very low
BRAIN REALISM	Low	Medium	High
TRAINING	Backpropagation (gradient descent)	Mixed approaches (BP, Hebbian, RL, local learning)	Often alternative learning methods (Hebbian, STDP, reinforcement learning)
BEST AT TODAY	LLMs, image generation, recommendation engines	Brain-modeling, adaptive systems research, temporal reasoning, sensor fusion	Robotics, edge AI, always-on sensors, low-power devices, real-time control
KEY STRENGTH	Highest accuracy for current AI tasks	More biologically plausible, better temporal processing	Ultra-low power, real-time response, continuous learning potential
KEY CHALLENGE	Energy intensive, not brain-like	Simulation overhead can be computationally expensive	Difficult programming model, immature tooling, smaller ecosystem
BEST FIT	Knowledge work, content generation, analysis	Research, adaptive systems, complex environments	Robotics, autonomous systems, edge devices, IoT

HOW IT WORKS, HIGH LEVEL

1 · Standard AI

Input ▼ Layers ▼ Layers ▼ Output

A neuron: weighted inputs → Σ summation → f() activation. Massive matrix math; neurons fire every clock cycle.

2 · Simulated Neuromorphic

Spike ▼ Membrane potential ▼ Spike ▼ Network dynamics

Input spikes → membrane integrates → threshold reached → spike out. Neurons fire when charge crosses threshold — still on standard hardware.

3 · Neuromorphic on Chip

Sensor ▼ Spikes ▼ Neuromorphic cores ▼ Spikes ▼ Action

Compute model: event → compute → idle → event. The chip only computes when events occur. Much closer to the brain; extremely low power.

CONTENT + OOUX ANALOGY

1 · Standard AI

Content → Vector embedding → Probability prediction

Predicts likely relationships.

2 · Simulated Neuromorphic

Object → Relationship activation → Temporal context → Meaning

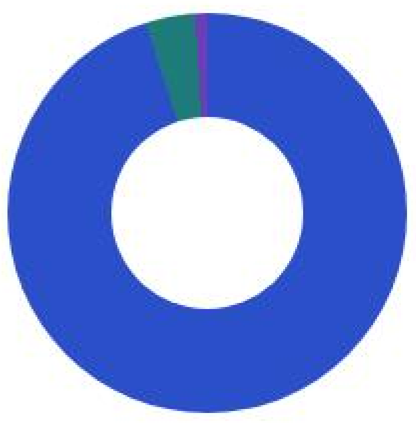
Activates relationships dynamically.

3 · Neuromorphic on Chip

Object detected → Relationships spike → Context activates → Action

Relationships continuously reshape behavior.

WHERE THE INDUSTRY IS TODAY — APPROXIMATE



- **Standard AI — ~95%.** Most commercial AI (GPT, Gemini, Claude, image generators) is standard AI running on GPUs.
- **Simulated Neuromorphic — ~4%.**
- **Neuromorphic Hardware — ~1%.**

Percentages are illustrative.

THE FUTURE — HYBRID COGNITIVE SYSTEMS

LLMs Language & reasoning + **Knowledge Graphs** Persistent memory + **Neuromorphic Architectures** Adaptive learning + **Neuromorphic Hardware** Energy-efficient execution

= More energy-efficient, continuously adaptive systems with persistent world models — not just next-token prediction.

WHY THE COMBINATION MATTERS

- **Energy efficiency** — Dramatically lower power for always-on intelligence.
- **Continuous adaptation** — Learns and adapts in real time.
- **Persistent world models** — Understand context over time, not just tokens.
- **Better robotics & edge AI** — Faster, safer, more capable real-world systems.
- **Resilient & scalable** — Works at the edge and in large-scale systems.

The long-term opportunity is not choosing one approach, but intelligently combining them—toward more intelligent, adaptive, and aligned AI, at the scale of the world.